Test Report - VxMark Thermals

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Summary

This investigation verified acceptable temperatures of components in VxMark in preparation for UL 62368-1 Safety Testing. All measured temperatures were within operating specifications for internal components and within safe touch limits for external surfaces.

Applicable VVSG Requirements

- 8.1-K – Eliminating hazards

Devices Under Test

- VxMark, v4.1, image v4.0.4
- SanDisk 64GB Ultra Flair USB3.0 USB drive

Purpose

This investigation documented temperatures around the VxMark unit both internally and externally while operating over time in different ambient temperatures. Measurement locations of particular interest were user-facing surfaces and temperatures at and around the SBC/SSD boards, which were previously shown to be the warmest parts of VxScan, as reported in *Test Report - VxScan Thermals* and *Test Report - VxScan Thermals with Updated Hardware/Software*. Temperatures were expected to be lower than in VxScan, within acceptable safe touch limits for user-facing surfaces, and within the operating specifications of internal components.

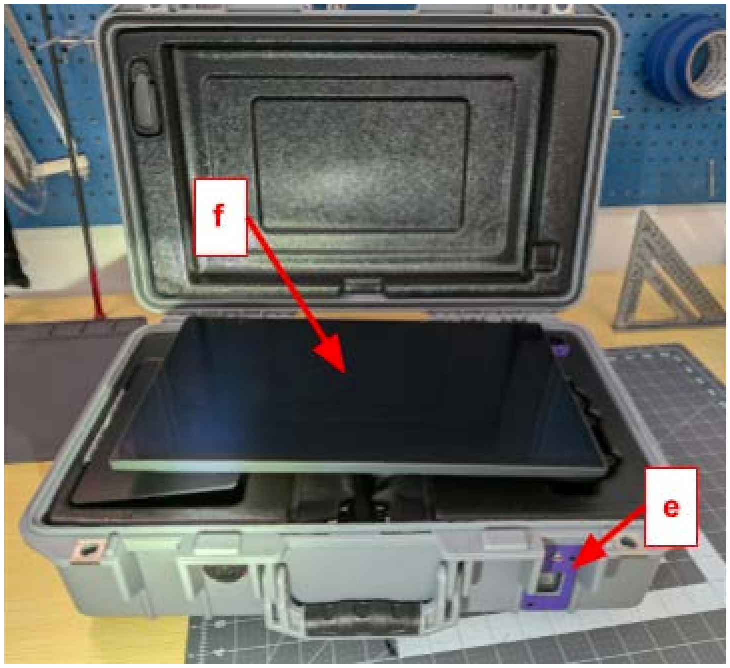
Materials

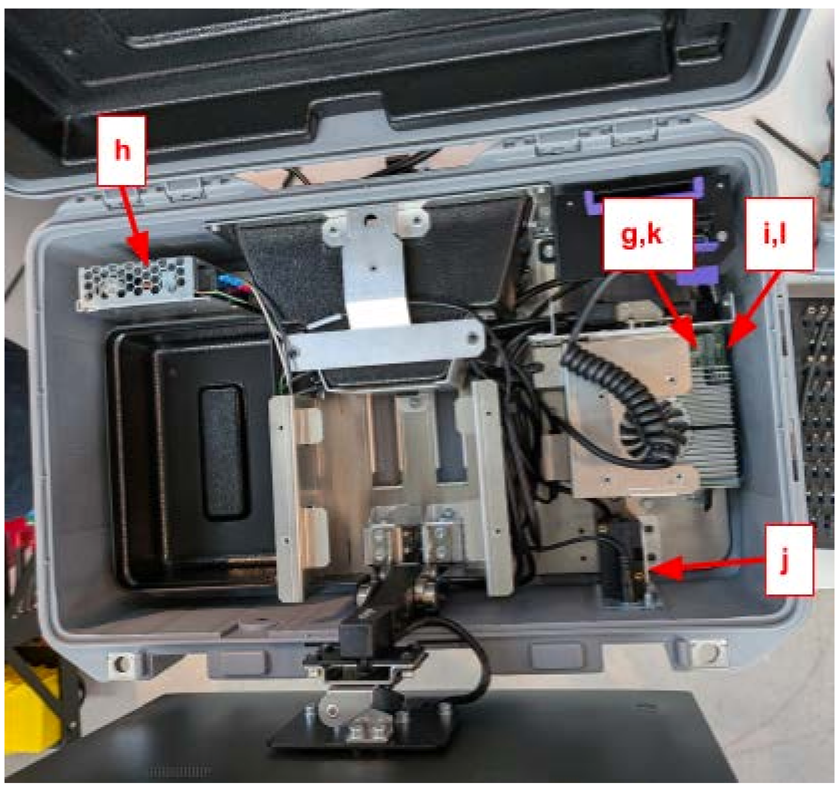
- VxMark units (MK-14-019 and MK-14-018)
    - Imaged with v4.0.4 production app
    - Set up on a table connected to printer.
- VxMark printer (HP4001dn)
- Tools for accessing VxMark internals
- Thermal camera: Topdon TC004

- Thermocouple system: 4-channel system, with K-type thermocouples
- Electrical tape, scissors
- Ambient thermometer
- Climate tent, table, space heater, humidifier

Procedures

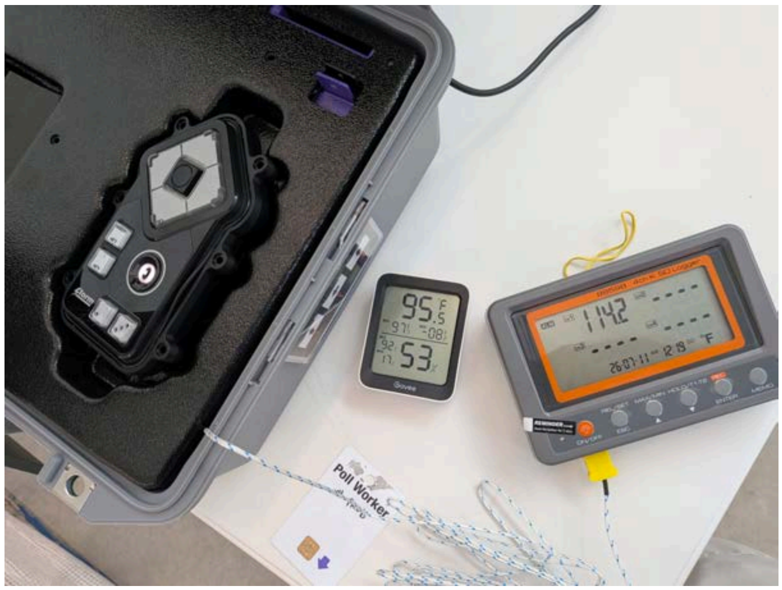
1. Key locations where temperatures were measured were defined below. Internal air temperatures were measured via thermocouple. Surface temperatures were measured via thermal camera. Selected locations are also pictured below.:
    a. External surface: right plastic panel above SBC, under ATI
    b. External surface: USB port
    c. External surface: rear plastic cover where cords exit
    d. External surface: left horizontal plastic panel rear to headphone storage
    e. External surface: Barcode scanner light
    f. External surface: Touchscreen, center
    g. Internal surface: SSD chip
    h. Internal surface: Power supply internals
    i. Internal surface: SBC front by bracket
    j. Internal surface: barcode scanner rear
    k. Internal air temperature: SSD chip
    l. Internal air temperature: SBC front by bracket



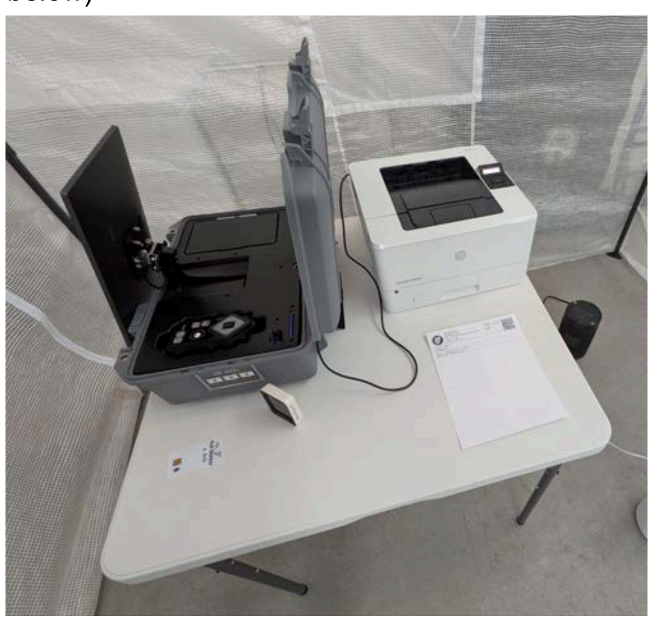


2. Temperatures were measured at these locations in typical (room temperature) and warm (95°F ambient) conditions, while on for over several hours. Printing a ballot occurred periodically every 20-30 minutes but was not expected to add much to the thermal load on the VxMark unit itself.

a. VxMark external user-facing surface temperatures were measured first using MK-14-019, without opening up the unit at all.
b. VxMark internal temperatures were then measured using MK-14-018, where the unit was opened up for access to the internals. The midplate screws were removed, and a thermocouple was affixed near the SBC and SSD boards. The midplate was then laid back down over the unit and left loose, so that it could be raised again to take measurements with the thermal camera. (Example photo below; note only one thermocouple wire is shown instead of two in this case.)



c. For warm temperature testing, the ambient temperature was raised to 95°F (35°C) and 55%RH around the test unit as per VVSG2.0 rated operational extremes, while following safety protocols for heating in the climate tent. (Photo of setup below)



d. Measurements were taken at these times and conditions, when possible:
    i. 30+ minutes idle, room temperature
    ii. 12+ hours idle (overnight), room temperature
    iii. 2 hours on, 95°F
    iv. 3 hours on, 95°F
    v. 4 hours on, 95°F
e. When measurements were complete, the system was turned off and stowed.

Results

Thermal measurements of VxMark were within component specifications and safe touch limits in all conditions. See Appendix A below for thermal measurement data.

The hottest temperatures were detected near the SBC board in warm ambient conditions, as expected. The maximum operating air temperature around the SBC was about 118°F (48°C) after the VxMark was on for about 4 hours in 95°F ambient conditions. This was still about 25°F cooler (13°C cooler) than rated operating temperature specifications and 11°F cooler (6°C cooler) than similar measurements around the SBC in VxScan in previous reports. The SBC board itself measured surface temperatures up to 151°F (66°C), and this is still below industry guidelines for chipmakers to keep temperatures under 85°C.

Worst-case user-facing surface temperatures were found under the ATI on the metal plate above the SBC, at about 123°F (51°C), in 95°F ambient conditions. This temperature is below the 140°F limit for safely touch metal as recommended by OSHA and ASTM C1055.

Conclusions

- Internal operating air temperatures in VxMark are within rated limits for components. Internal electronic components run cool enough not to expect any performance throttling.
- External user-facing temperatures in VxMark are within safe touch limits.

Appendix A: VxMark thermal measurements

Thermals with ADATA SSD, updated hardware and software March 2026				Temperature measurements (degrees F)						
				Room temperature ambient		Warm ambient				
Location	**User-facing?**	**Type**	**Measurement tool**	**Idle 30+ min**	**Idle overnight**	**On 2 hours, 95F, 55%RH**	**On 3 hours, 95F, 55%RH**	**On 4 hours, 95F, 55%RH**		**Spec: Max operating (F)**
right plastic panel above SBC, under ATI	yes	surface	thermal imager	101.7	100.6	119.5	122.8	117.2		140 (OSHA)
USB port	yes	surface	thermal imager	97.8	92.5	115.0	119.5	115.0		140 (OSHA)
rear plastic cover where cords exit	yes	surface	thermal imager	84.5	87.0	110.1	111.2	106.2		140 (OSHA)
left horizontal plastic panel rear to headphone storage	yes	surface	thermal imager	85.5	85.6	108.6	110.4	105.8		140 (OSHA)
Barcode scanner light	yes	surface	thermal imager	92.2	93.3	116.5	119.2	116.1		140 (OSHA)
Touchscreen, center	yes	surface	thermal imager	82.1	81.6	105.4	105.4	100.4		140 (OSHA)
SSD chip	no	surface	thermal imager	129.6	not measured	161.7	154.7	149.3		n/a
Power supply internals	no	surface	thermal imager	122.9	not measured	146.1	145.6	141.9		158 (spec)
SBC front by bracket, chip	no	surface	thermal imager	118.5	not measured	151.1	144.9	139.5		n/a
barcode scanner rear	no	surface	thermal imager	98.6	not measured	131.4	125.9	121.5		140F (spec)
SSD chip, ambient	no	air, surrounding	thermocouple	not measured		111.1	not measured			185 (spec)
SBC front by bracket, ambient	no	air, surrounding	thermocouple	not measured		114.2	117.1	116.8		140F (spec)